Some Queuing Formulas

In all cases, $\rho = \lambda/(s\mu)$.

Model	P_0	P_n
M/M/1	$1-\rho$	$(1-\rho)\rho^n$
M/M/s	$\left(\sum_{n=0}^{s-1}\left(\frac{\lambda}{\mu}\right)^n\frac{1}{n!}+\left(\frac{\lambda}{\mu}\right)^s\frac{1}{s!}\frac{1}{1-\lambda/(s\mu)}\right)^{-1}$	$\begin{cases}\frac{\lambda^n}{n!\mu^n}P_0, & n\le s\\ \frac{\lambda^n}{s!s^{n-s}\mu^n}P_0, & n>s\end{cases}$
M/M/1/K	$\begin{cases}\frac{1-\rho}{1-\rho^{K+1}}, & \lambda\neq\mu\\ \frac{1}{K+1}, & \lambda=\mu\end{cases}$	$\begin{cases}\rho^nP_0, & \lambda\neq\mu\\ \frac{1}{K+1}, & \lambda=\mu\end{cases}$
M/M/1 (finite population)	$\left(\sum_{n=0}^{N}\frac{N!}{(N-n)!}\rho^n\right)^{-1}$	$\frac{N!}{(N-n)!}\rho^nP_0$
M/G/1	$1-\rho$	N/A

Model	L	L_q
M/M/1	$\frac{\lambda}{\mu-\lambda}$	$\frac{\lambda^2}{\mu(\mu-\lambda)}$
M/M/s	$L_q+\frac{\lambda}{\mu}$	$\frac{\rho\lambda^s}{s!(1-\rho)^2\mu^s}P_0$
M/M/1/K	$\begin{cases}\frac{\rho}{1-\rho}-\frac{(K+1)\rho^{K+1}}{1-\rho^{K+1}}, & \lambda\neq\mu\\ \frac{K}{2}, & \lambda=\mu\end{cases}$	$L-(1-P_0)$
M/M/1 (finite population)	$N-\frac{\mu}{\lambda}(1-P_0)$	$N-\frac{\mu+\lambda}{\lambda}(1-P_0)$
M/G/1	$\rho+L_q$	$\frac{\lambda^2\sigma^2+\rho^2}{2(1-\rho)}$
M/D/1	$\rho+L_q$	$\frac{\rho^2}{2(1-\rho)}$
M/E_k/1	$\rho+L_q$	$\frac{1+k}{2k}\frac{\lambda^2}{\mu(\mu-\lambda)}$